

Question#1

```
#include<stdio.h>
int main(){
    int number;
    double factorial=1;
    printf("Enter a number whose factorail you want: ");
    scanf("%d",&number);
    for(int i=1;i<=number;i++){
        factorial =i*factorial;
    }
    printf("Factorial of %d is %lf ",number,factorial);
    return 0;
}
```

Question#2

```
#include <stdio.h>
int main()
{
    int number, sum = 0;
    printf("Enter a five digit number: ");
    scanf("%d", &number);
    int temp = number;
    int n = number;
    while (n > 0)
    {
        sum += n % 10;
        n = n / 10;
    }
    if (sum % 2 == 0)
    {
        int prime = 1;
        if (sum < 2)
        {
            prime = 0;
        }
        else
        {
            for (int i = 2; i <= sum / 2; i++)
            {
                if (sum % i == 0)
                {
                    prime = 0;
                    break;
                }
            }
        }
        if (prime){
            printf("Even sum %d and it is Prime\n", sum);
        }
        else{
            printf("Even sum %d but it is Not Prime\n", sum);
        }
    }
    else
    {
        int original = temp, reversed = 0, remainder;
        while (temp > 0)
        {
            remainder = temp % 10;
            reversed = reversed * 10 + remainder;
            temp /= 10;
        }
        if (original == reversed){
            printf("Odd sum %d and number %d is Palindrome\n", sum, sum);
        }
        else{
            printf("Odd sum %d and number %d is Not Palindrome\n", sum, sum);
        }
    }
    return 0;
}
```

Questio#3

For Code

```
#include<stdio.h>
int main(){
    int number;
    printf("Enter any odd number:");
    scanf("%d",&number);
    for(int i=1;i<=number;i++){
        for(int space=1;space<=number-i;space++){
            printf(" ");
        }
        for(int j=1;j<=i*2-1;j++){
            printf("*");
        }
        printf("\n");
    }
    for (int i = number - 1; i >= 1; i--) {
        for (int space = 1; space <= number - i; space++){
            printf(" ");
        }
        for (int j = 1; j <= 2 * i - 1; j++){
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

While Code

```
#include <stdio.h>
int main() {
    int number;
    printf("Enter any odd number: ");
    scanf("%d", &number);
    int i = 1;
    while (i <= number) {
        int space = 1;
        while (space <= number - i) {
            printf(" ");
            space++;
        }
        int j = 1;
        while (j <= i * 2 - 1) {
            printf("*");
            j++;
        }
        printf("\n");
        i++;
    }
    i = number - 1;
    while (i >= 1) {
        int space = 1;
        while (space <= number - i) {
            printf(" ");
            space++;
        }
        int j = 1;
        while (j <= 2 * i - 1) {
            printf("*");
            j++;
        }
        printf("\n");
        i--;
    }
    return 0;
}
```

Question#4

```
#include <stdio.h>

int main() {
    char username[6];
    char password[7];
    char login_user[6];
    char login_pass[7];
    int valid_username = 0;
    int valid_password = 0;
    int i;
    int len;
    while (valid_username == 0) {
        printf("Enter username (exactly 5 alphabets): ");
        scanf("%s", username);
        // Length check
        len = 0;
        while (username[len] != '\0') {
            len++;
        }
        if (len != 5) {
            printf("Invalid: Username must be exactly 5 characters.\n");
            continue;
        }
        valid_username = 1;
        for (i = 0; i < 5; i++) {
            char ch = username[i];
            if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {
                continue;
            } else {
                printf("Invalid: Username must contain only alphabets (A-Z, a-z).\n");
                valid_username = 0;
                break;
            }
        }
    }

    while (valid_password == 0) {
        printf("Enter password (exactly 6 characters, at least 1 digit, 1 uppercase, 1 lowercase): ");
        scanf("%s", password);

        // Length check
        len = 0;
        while (password[len] != '\0') {
            len++;
        }
        if (len != 6) {
            printf("Invalid: Password must be exactly 6 characters.\n");
            continue;
        }

        // Check flags and loop
        int has_digit = 0;
        int has_upper = 0;
        int has_lower = 0;
        valid_password = 1;

        for (i = 0; i < 6; i++) {
            char ch = password[i];

            // Check for digit
            if (ch >= '0' && ch <= '9') {
                has_digit = 1;
                continue;
            }

            // Check for uppercase
            if (ch >= 'A' && ch <= 'Z') {
                has_upper = 1;
                continue;
            }
        }
    }
}
```

```

    }

    // Check for Lowercase
    if (ch >= 'a' && ch <= 'z') {
        has_lower = 1;
        continue;
    }
}
if (has_digit == 0 || has_upper == 0 || has_lower == 0) {
    printf("Invalid: Password must have at least 1 digit, 1 uppercase, and 1 lowercase.\n");
    valid_password = 0;
}
}
printf("Account Created Successfully\n");

printf("Login:\n");
printf("Enter username: ");
scanf("%s", login_user);
printf("Enter password: ");
scanf("%s", login_pass);

int user_len = 0;
while (login_user[user_len] != '\0') {
    user_len++;
}
int match_username = (user_len == 5);
if (match_username) {
    for (i = 0; i < 5; i++) {
        if (login_user[i] != username[i]) {
            match_username = 0;
            break;
        }
    }
}
}

```

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int pass_len = 0;
while (login_pass[pass_len] != '\0') {
    pass_len++;
}
int match_password = (pass_len == 6);
if (match_password) {
    for (i = 0; i < 6; i++) {
        if (login_pass[i] != password[i]) {
            match_password = 0;
            break;
        }
    }
}

if (match_username && match_password) {
    printf("Welcome %s, you are now logged in.\n", username);
} else {
    printf("Login failed: Invalid username or password.\n");
}

return 0;
}

```

Question#5

```
#include <stdio.h>
int main() {
    int n;
    int i, j;
    printf("Enter the number of rows: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Invalid: Number of rows must be positive.\n");
        return 1;
    }
    if (n >= 1) {
        printf("1\n");
    }
    if (n >= 2) {
        printf("1 1\n");
        long long prev_second_last = 1;
        long long prev_last = 1;
        for (i = 3; i <= n; i++) {
            long long num1 = prev_second_last + prev_last;
            printf("%lld ", num1);
            long long num2 = num1 + prev_last;
            printf("%lld ", num2);
            long long curr_second_last = num1;
            long long curr_last = num2;
            for (j = 3; j <= i; j++) {
                long long numj = curr_last + curr_second_last;
                printf("%lld ", numj);
                curr_second_last = curr_last;
                curr_last = numj;
            }
            printf("\n");
            prev_second_last = curr_second_last;
            prev_last = curr_last;
        }
    }
    return 0;}
```

Question#6

```
#include <stdio.h>
int main() {
    int n;
    int i, j;
    printf("Enter the number of rows positive integer: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Invalid: Number of rows must be positive.\n");
        return 1;
    }
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) {
            printf("%d", j);
        }
        for (j = i - 1; j >= 1; j--) {
            printf("%d", j);
        }
        printf("\n");
    }
    return 0;
}
```

Question#7

```
#include <stdio.h>
int main() {
    char sentence[500];
    int word_start = 0, word_end = 0, num_words = 0, j = 0, k = 0, i=0;
    printf("Enter a sentence : ");
    scanf("%s", sentence);
    while (sentence[i] != '\0') {
        if (sentence[i] == ' ') {
            num_words++;
        }
        i++;
    }
    num_words++;
    if (num_words == 0) num_words = 1;
    printf("Reversed word order: ");
    int current_word = num_words;
    while (current_word > 0) {
        word_start = 0;
        int words_found = 0;
        j = 0;
        while (words_found < current_word - 1 && sentence[j] != '\0') {
            if (sentence[j] == ' ') {
                words_found++;
                word_start = j + 1;
            }
            j++;
        }
        word_end = word_start;
        while (sentence[word_end] != ' ' && sentence[word_end] != '\0') {
            word_end++;
        }
        for (k = word_start; k < word_end; k++) {
            printf("%c", sentence[k]);
        }
        if (current_word > 1) {
            printf(" ");
        }
        current_word--;
    }
    printf("\n");
    return 0;
}
```


Question#8

```
#include <stdio.h>
int main() {
    long long number, original;
    int digits = 0;
    long long sum = 0;
    int digit;
    long long temp;
    long long power_value;
    printf("\nCheck Armstrong Number\n\n");
    printf("\nEnter a non-negative integer:");
    scanf("%lld", &number);
    if (number < 0) {
        printf("\nInvalid: Negative numbers are not Armstrong numbers.\n");
        return 1;
    }
    if (number == 0) {
        printf("0 is not considered an Armstrong number.\n");
        return 0;
    }
    temp = number;
    while (temp > 0) {
        digits++;
        temp /= 10;
    }
    temp = number;
    while (temp > 0) {
        digit = temp % 10;
        power_value = 1;
        for (int i = 1; i <= digits; i++) {
            power_value *= digit;
        }
        sum += power_value;
        temp /= 10;
    }
    if (sum == original) {
        printf("Yes, %lld is an Armstrong number!\n", original);
    } else {
        printf("No, %lld is not an Armstrong number.\n", original);
    }
    return 0;
}
```

Question#9

```
#include <stdio.h>
int main() {
    long long number, original, abs_num;
    int digit;
    int digits = 0;

    printf("Enter a number: ");
    scanf("%lld", &number);
    original = number;
    abs_num = (original > 0 ? original : -original);
    long long temp = abs_num;
    while (temp > 0) {
        digits++;
        temp /= 10;
    }
    if (original == 0) digits = 1;
    int i, k;

    for (i = 1; i <= digits; i++) {
        long long power = 1;
        for (k = 1; k <= (digits - i); k++) {
            power *= 10;
        }
        digit = (abs_num / power) % 10;
        printf("On %d position digit is %d\n", i, digit);
    }

    return 0;
}
```

Question#10

```
#include <stdio.h>

int main()
{
    int number, sum = 0;
    int position = 0;
    printf("Enter a number: ");
    scanf("%d", &number);
    while (number > 0)
    {
        int digit = number % 10;

        if (position % 2 != 0)
        {
            sum += digit;
        }

        number = number / 10;
        position++;
    }
    printf("Sum of odd position digits: %d\n", sum);
    return 0;
}
```